SUPPLY CHAIN PERFORMANCE INVESTIGATION

Year-over-year findings, diagnostic risks, and management actions

PREPARED FOR
Supply Chain Management

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DATA SOURCE
DataCo Supply Chain — cleaned star-schema exports and Power BI dashboards

PERIOD COVERED
2015–2017

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PURPOSE

Translate dashboard signals into evidence-based management priorities, while distinguishing confirmed observations from questions that still require data validation.

Executive Summary

The dashboards show a business with persistent delivery problems and a weakening commercial picture in 2017. From 2015–2016 to 2016–2017, headline sales fell from $12.30M to $11.81M, while the number of orders increased to approximately 22K. Quantity sold and average order value declined, indicating that order volume alone did not sustain sales performance.

Delivery reliability remained the clearest operational concern. The late-delivery rate was 55.12% in 2016 and 54.54% in 2017. First Class and Second Class had the highest reported late-delivery rates in both years. Although the overall rate improved slightly in 2017, the dashboard still indicates that more than half of orders were late.

Profitability was more resilient than revenue in 2017: total profit was approximately $1.30M and profit margin rose to 12.29%. At the same time, the customer dashboard reported a 48.3% increase in distinct customers and a 35.3% decline in revenue per customer. These shifts deserve investigation, not automatic interpretation as a change in customer behavior.

Priority diagnostic before management action

The dashboard shows a sharp decline in monthly sales in November and December 2017 while order counts remain comparatively high. The existing data review also flags an unusual pattern of one item row and one unit per order in those months. Reconcile this against the raw source and cleaned exports before treating it as a genuine change in purchasing behavior.

Key findings at a glance

Area	Evidence in the dashboard	Management implication
Commercial	2017 sales: $11.81M (-4.0%); orders: 22K (+4.8%); quantity: 106K (-22.7%).	Investigate lower sales per order and the late-2017 pattern.
Delivery	Late-delivery rate: 54.54%; about 12K late orders in 2017.	Validate status/date logic, then prioritize shipping-mode drivers.
Profitability	2017 profit: $1.30M (-0.5%); margin: 12.29% (+0.44 percentage points).	Protect margin as sales soften.
Customer value	Distinct customers: about 15K (+48.3%); revenue/customer: $700.11 (-35.3%).	Check metric consistency and investigate value per customer.

Interpretation rule: the report describes patterns in the supplied dashboards. Where the pattern may be affected by metric definitions, model relationships, or data quality, it is marked for verification rather than presented as a proven cause.

1. Scope and Methodology

1.1 Objective and comparison periods

This report summarizes the four Power BI pages—Executive Overview, Supply Chain Performance, Product & Department Analysis, and Customer & Market Analysis—for two year-over-year comparisons: 2015 vs. 2016 and 2016 vs. 2017. The purpose is to identify decision-relevant signals and propose a disciplined follow-up plan.

1.2 Data and analytical approach

The project uses cleaned DataCo Supply Chain CSV exports organized into a star-schema-style model, including separate order-level and order-item-level fact tables. That structure makes it important to match every metric to the correct grain: distinct order IDs for order counts, order-item records for quantities and item-level revenue, and a consistent customer definition for customer counts.

- Compare the headline KPI values and the dashboard's year-over-year indicators.
- Review revenue contributions by market, department, category, product, and customer segment.
- Compare reported late-delivery rates and shipping-time averages by shipping mode and market.
- Separate observed results from potential explanations and specify the validation needed before action.

1.3 Metric interpretation

Measure	How to read it in this report
Total Sales vs. Net Revenue	These are separate dashboard measures and should not be used interchangeably. Keep comparisons within the same measure and visual.
Late-delivery rate	A reported outcome rate. A high value identifies a pattern to investigate, but does not prove that the shipping mode itself caused the delay.
Shipping variance	The dashboard reports the difference between average actual and scheduled shipping days. This is not necessarily the same as the average of order-level delivery delays.
Year-over-year indicators	Follow the dashboard's displayed change, but confirm whether a percentage label represents a relative change or a percentage-point difference.

Analytical limitation

This is descriptive and diagnostic analysis, not a causal study. Market shifts, late-delivery rates, and customer-value changes should be reconciled to the underlying records and metric definitions before management allocates budget or changes policy.

2. Findings and Analysis

A. 2015 vs. 2016

2.1 Executive Overview

Business performance remained broadly stable in 2016, with sales at $12.30M. The year-over-year dashboard indicators show small declines in sales (-0.3%), orders (-0.2%), quantity sold (-0.8%), and average order value (-0.1%). These changes suggest modest softening rather than a major headline-level contraction.

USCA ($5.04M) and Pacific Asia ($4.45M) were the leading markets in the sales visual, followed by Africa ($2.05M) and Europe ($0.76M). Fan Shop was the leading department on the Executive Overview page, at approximately $6.2M. Revenue was therefore concentrated in a small number of markets and one leading department.

The late-delivery rate was 55.12%, making service reliability the main operational concern in the overview. First Class and Second Class had particularly high reported rates—95.34% and 76.98%, respectively.

2.2 Supply Chain Performance

The dashboard reports approximately 11K late-delivery orders. Average actual shipping time was 3.49 days versus 2.92 scheduled days, a 0.57-day gap between the two reported averages. First Class (95.34%) and Second Class (76.98%) had the highest late-delivery rates; Same Day (43.61%) and Standard Class (38.57%) were lower.

What management should take from this

Delivery reliability is a clear priority for investigation. Before changing shipping policy, validate the late-delivery flag against shipment dates and delivery status, then compare rates by order profile, market, and shipping mode.

A. 2015 vs. 2016 (continued)

2.3 Product & Department Analysis

The product dashboard reports $12.30M in sales, $11.06M in net revenue, and $1.31M in total profit. Profit margin was 11.85%, with the year-over-year indicators showing a 0.7% decline in total profit and a small decrease in margin. This points to a need to review profitability alongside revenue, rather than treating high sales as a complete measure of product performance.

On the Product & Department page, Fan Shop led department net revenue at approximately $5.6M, followed by Apparel ($2.4M) and Golf ($1.5M). Fishing led the category chart at $2.3M, followed by Cleats ($1.5M) and Camping & Hiking ($1.3M). Note that the department value differs from the Executive Overview visual because the pages display different measures; use the exact measure label when comparing values.

2.4 Customer & Market Analysis

The customer dashboard reported approximately 10K distinct customers and $1,081.59 in revenue per customer, with the KPI indicator showing a 1.2% decline in revenue per customer. Consumers were the largest customer segment, contributing about $5.8M in net revenue and $0.7M in profit, followed by Corporate and Home Office.

USCA generated the highest market profit at approximately $560K, followed by Pacific Asia at $448K. The segment-market matrix also indicates comparatively weaker margins for some Pacific Asia customer-segment combinations. These figures can guide a deeper margin review, but should be considered alongside the same segment definitions and the corresponding revenue measures.

2016 takeaway

Revenue was concentrated in Fan Shop, USCA, and Pacific Asia, while delivery reliability and margin management remained important improvement areas. The figures describe the 2016 dashboard; they do not, by themselves, establish the cause of year-over-year changes.

B. 2016 vs. 2017

3.1 Executive Overview

The 2017 dashboard shows a weaker commercial picture: Total Sales declined by 4.0% to $11.81M, while orders increased by 4.8% to approximately 22K. Quantity sold fell by 22.7% to about 106K, and average order value declined by 8.4% to $540.04. In short, more orders did not translate into stronger sales, and the large reduction in quantity warrants a closer look at units per order and revenue per order.

The sales-by-market visual shows LATAM ($5.21M) and Europe ($5.20M) as the leading markets, while Pacific Asia contributed $1.13M, Africa $0.25M, and USCA about $0.02M. This is a striking shift from the 2016 dashboard, where USCA and Pacific Asia were the leading markets. It should be reconciled to the raw data, geography mapping, year filters, and measure definitions before being treated as a genuine market transition.

Fan Shop remained the leading department in the Executive Overview visual, at approximately $4.7M. The late-delivery rate eased to 54.54%, but still affected more than half of orders.

3.2 Supply Chain Performance

The supply-chain page reports approximately 12K late orders. Average actual shipping time was 3.50 days versus 2.94 scheduled days, a 0.56-day gap between reported averages. First Class recorded a 94.50% late-delivery rate and Second Class 76.43%. Same Day was 50.37%, while Standard Class was lowest at 37.74%. By market, Pacific Asia was highest at 55.09%, followed by Europe (54.81%) and LATAM (54.25%); USCA was lowest at 43.90%.

Operational interpretation

The overall late-delivery rate improved by about 0.58 percentage points from 2016 to 2017, but it remains high. First and Second Class warrant investigation, while the very large market mix shift and the end-of-year sales pattern should be validated before operational causes are assigned.

B. 2016 vs. 2017 (continued)

3.3 Product & Department Analysis

The product dashboard reports $11.81M in Total Sales, $10.61M in Net Revenue, and $1.30M in profit. Compared with the prior year, sales and net revenue were each down 4.0%, while profit declined by only 0.5%. Profit margin improved by 0.44 percentage points to 12.29%, indicating that margin held up better than revenue.

Fan Shop remained the largest department on the Product & Department page at approximately $4.3M in net revenue, followed by Apparel ($2.3M) and Golf ($1.1M). Fishing was the top category at $1.71M, followed by Cleats ($1.08M) and Camping & Hiking ($1.04M). The dashboard reported 118 distinct products sold, up 118.5%. The product revenue-versus-margin visual also suggests that product performance is uneven: several products generate substantial revenue, while many contribute much less and some have negative margins.

Management should protect availability for important products but use product-level margin and contribution—not revenue ranking alone—to guide assortment, promotion, and inventory decisions.

3.4 Customer & Market Analysis

The customer dashboard reports approximately 15K distinct customers (+48.3%) but revenue per customer of $700.11 (-35.3%). Net revenue was $10.61M and total profit $1.30M. Consumers were the largest segment at about $5.5M in net revenue, followed by Corporate ($3.2M) and Home Office ($1.9M). Customer growth did not translate into proportional revenue growth, although the movement should be checked against the definition and year filter used for distinct customers.

LATAM generated the highest reported market profit at approximately $581K, closely followed by Europe at $555K. Pacific Asia was much lower at $133K, Africa at $30K, and USCA at about $4K. These profit rankings differ sharply from the prior-year dashboard, reinforcing the need to reconcile market assignment and measure logic before making resource-allocation decisions.

2017 takeaway

Profit margin improved despite lower sales, but customer value and market-profit distribution need investigation. Resolve the data consistency questions before treating the changes as definitive commercial or customer-behavior trends.

4. Conclusions and Recommendations

4.1 Conclusion

Across the two comparisons, the strongest recurring operational signal is late delivery: reported rates stayed above 54%, with First Class and Second Class consistently highest. The commercial picture weakened in 2017 as sales, quantity sold, and average order value declined despite an increase in orders. Profitability was more resilient, with the 2017 profit margin improving to 12.29%. Customer count increased sharply, but revenue per customer fell.

Several dashboard movements are unusually large—notably the November–December 2017 pattern and the shift in revenue and profit across markets. The responsible conclusion is not to assume a single business cause, but to verify data structure, metric definitions, filters, and model relationships before management commits to interventions.

4.2 Recommended management priorities

1. Validate the data before acting on unusual shifts.

Reconcile November and December 2017 order-item rows to the raw source. Check row counts, units per order, net revenue per order, date fields, market mapping, year filters, and relationships between order-level and order-item-level facts.

2. Improve delivery reliability.

Once the delivery outcome is validated, investigate First Class and Second Class first. Compare scheduled and actual dates by market, order profile, carrier (where available), and month. Set service targets only after establishing a trusted baseline.

3. Restore revenue per order and customer value.

If the late-2017 decline is confirmed as genuine, examine units per order, revenue per order, product mix, and customer segment. Test bundle offers, cross-selling, and targeted retention actions only after identifying which segments or baskets account for the decline.

4. Conclusions and Recommendations (continued)

4. Protect profitability through product-level decisions.

Use net revenue, profit, and margin together to identify products that combine scale with healthy contribution. Review negative-margin and high-revenue/low-margin items before changing prices, promotions, or inventory commitments.

5. Reconcile market performance before reallocating investment.

Validate the sharp change in market distribution between 2016 and 2017. Once confirmed, compare each market's net revenue, profit, margin, customer mix, and delivery performance using consistent definitions.

6. Make KPI monitoring part of the operating rhythm.

Review sales, net revenue, orders, quantity, revenue per customer, profit margin, late-delivery rate, and shipping variance monthly. Assign an owner to each metric and document thresholds, definitions, and follow-up actions.

Suggested implementation plan

Timing	Owner (proposed)	Priority actions	Evidence of completion
First 30 days	Data analyst / BI owner	Reconcile Q4 2017 order-item rows, metric definitions, customer counts, market allocation, and relationship paths.	Raw-to-clean row reconciliation; documented metric dictionary; validated year filters.
Days 31–60	Supply chain operations	Validate delivery status against shipment dates; diagnose First Class and Second Class delays by market and month.	Agreed late-delivery baseline; ranked delay drivers; proposed service targets.
Days 61–90	Commercial + finance	Analyze revenue per order/customer and product margins; select targeted experiments or assortment reviews.	Segment/product opportunity list; experiment plan; agreed revenue and margin KPIs.

Validation checklist before final sign-off

- Confirm that Total Sales and Net Revenue definitions are documented and comparable across years.
- Verify the November–December 2017 one-row/one-unit-per-order pattern against raw records.
- Reconcile the sharp change in market revenue and market profit between 2016 and 2017.
- Confirm distinct-customer calculations use the same grain, year context, and filter behavior.
- Check that late-delivery status and risk flags are not being counted as independent evidence if they encode the same outcome.

Final management message

Prioritize data validation and delivery reliability first. Then use trusted customer-value and product-margin measures to restore sales quality without sacrificing the profitability improvement reported in 2017.